

Project Design Phase-I
Proposed Solution Template

Date	07 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - Insurance Fraud Detection Using Machine Learning
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Insurance fraud is a major issue where some customers submit false or exaggerated insurance claims to gain money. Detecting such fraud manually is difficult because insurance companies process a large number of claims daily. Traditional methods are time-consuming and not always accurate in identifying fraud patterns. This leads to financial losses for insurance companies. Therefore, a machine learning-based system is needed to automatically analyze claim data and detect fraudulent claims efficiently.
2.	Idea / Solution description	The proposed solution is to develop a machine learning-based system that can automatically detect fraudulent insurance claims. The system will analyze insurance claim data such as customer details, policy information, and claim amount. Machine learning algorithms will be trained on historical data to identify patterns of fraudulent behavior. The model will predict whether a claim is fraudulent or genuine. This will help insurance companies detect fraud quickly, reduce financial losses, and improve decision-making.
3.	Novelty / Uniqueness	The uniqueness of this project lies in using machine learning algorithms to automatically detect fraudulent insurance claims instead of relying only on manual verification. The system analyzes multiple claim features and identifies hidden fraud patterns from the data. It provides faster and more accurate fraud detection. This helps insurance companies reduce losses and improve claim verification efficiency.
4.	Social Impact / Customer Satisfaction	This project helps insurance companies detect fraudulent claims quickly and reduce financial losses. It ensures that genuine customers receive their claim approvals faster without unnecessary delays. By preventing fraud, it helps maintain fairness in the insurance system. It also increases trust and satisfaction among customers. Overall, the system improves

		transparency and efficiency in the insurance claim process.
5.	Business Model (Revenue Model)	The system can be offered to insurance companies as a software solution or subscription-based service. Insurance companies can pay a monthly or yearly fee to use the fraud detection system. The model helps companies save money by reducing fraudulent claims. Additional revenue can be generated through system maintenance, updates, and data analysis services. This creates a sustainable and scalable business model
6.	Scalability of the Solution	Scalability of the Solution The proposed system is scalable and can handle a large volume of insurance claim data from multiple users. As more data becomes available, the machine learning model can be retrained to improve accuracy. The system can also be integrated with different insurance platforms and expanded to detect fraud in health, vehicle, and property insurance. This makes the solution adaptable for large organizations and growing datasets